Mercato: Full-Stack E-Commerce App

A complete online store with customer accounts, product search and filters, cart, Stripe card payments and an admin dashboard. Built with React, Node.js, Express and MySQL.

Live demo: https://mercato-pied.vercel.app

> The demo runs on free hosting. If nothing loads at first, wait about a minute and refresh. The server wakes up after being idle.

Test payment: use card `4242 4242 4242 4242`, any future expiry date and any CVC. Stripe is in test mode, so no real money is charged.

Features :

Customers :

- Sign up and log in (JWT authentication, passwords hashed with bcrypt)
- Browse 660 products across 6 categories
- Search, filter by category, price range and rating, sort, and page through results
- Cart that is saved in the browser, with quantity changes and a live total
- Checkout with Stripe, then a payment confirmation page
- My orders page, with a cancel option for unpaid orders (stock is returned)

Admin :

- Add, edit and delete products, with search and paging for large catalogs
- Low-stock highlight so empty shelves are easy to spot
- Create categories
- View all orders and update their status (placed, shipped, delivered, cancelled)

What makes it solid

- Prices are calculated on the server. The browser only sends product IDs and quantities, so totals cannot be tampered with.
- Stock is handled in database transactions with row locks, so two people cannot buy the last item at the same time.
- Payments are verified with Stripe. An order is marked as paid only after the server confirms the PaymentIntent succeeded with Stripe (a webhook handler is included too).
- Role-based access. Admin routes are protected, and users can only see their own orders.

Tech stack :

Part	Technology

```
| Frontend | React, Vite, Tailwind CSS, React Router |
| Backend | Node.js, Express |
| Database | MySQL with Sequelize |
| Auth | JWT, bcrypt |
| Payments | Stripe (PaymentIntents, test mode) |
| Hosting | Vercel (frontend), Render (API), Aiven (MySQL) |
```

How it fits together :

```
React app (Vercel)  ->  Express API (Render)  ->  MySQL (Aiven)
                                  |
                              Stripe
```

Project structure :

```
mercato-store
├─ backend
│   ├─ config          database connection
│   ├─ controllers     auth, products, orders, payments
│   ├─ middleware      login and admin checks
│   ├─ models          Users, Categories, Products, Orders, OrderItems
│   ├─ routes
│   ├─ seed.js         adds 6 categories and 660 sample products
│   ├─ make-admin.js turns a registered user into an admin
│   └─ server.js
└─ frontend
    └─ src             pages, components, cart and auth context
```

API overview :

```
| Method | Endpoint | Access |
|---|---|---|
| POST | `/api/auth/register`, `/api/auth/login` | Public |
| GET | `/api/auth/me` | Logged in |
| GET | `/api/products`, `/api/products/:id`, `/api/products/categories`
| Public |
| POST, PUT, DELETE | `/api/products`, `/api/products/:id` | Admin |
| POST | `/api/orders` | Logged in |
| GET | `/api/orders/my`, `/api/orders/:id` | Logged in |
| PUT | `/api/orders/:id/cancel` | Logged in |
| GET | `/api/orders` | Admin |
| PUT | `/api/orders/:id/status` | Admin |
| POST | `/api/payments/create-intent`, `/api/payments/confirm` | Logged
in |
| POST | `/api/payments/webhook` | Stripe |

Product list filters: `search`, `category`, `minPrice`, `maxPrice`,
`minRating`, `sort`, `page`, `limit`.
```

Run it on your computer :

You need Node.js 18 or newer and MySQL.

Windows, the easy way: double-click `START.bat`. It creates the settings files, installs packages and starts both servers.

Manual setup :

bash

- Backend

```
cd backend
cp .env.example .env        # fill in your MySQL password and Stripe test key
npm install
npm run dev                 # runs on http://localhost:5000
```

- Frontend (new terminal)

```
cd frontend
cp .env.example .env        # add your Stripe publishable key
npm install
npm run dev                 # runs on http://localhost:5173
```

The database is created automatically on the first run.

Add sample data and an admin:

bash

```
cd backend
node seed.js --fresh                  # 660 products in 6 categories
node make-admin.js you@example.com    # register on the site first
```

Environment variables :

Backend: `PORT`, `DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASS`, `DB_SSL`, `JWT_SECRET`, `STRIPE_SECRET_KEY`, `STRIPE_WEBHOOK_SECRET`, `CURRENCY`, `CLIENT_URL`

Frontend: `VITE_API_URL`, `VITE_STRIPE_PUBLISHABLE_KEY`, `VITE_CURRENCY`

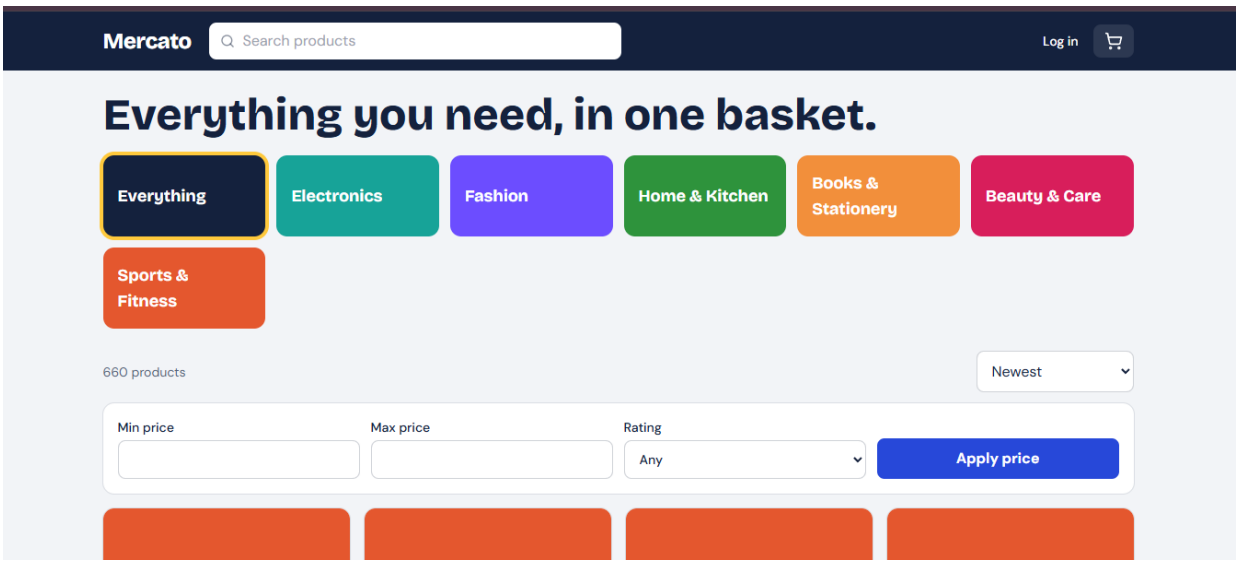
Real keys and passwords are never committed to this repository.

Deployment :

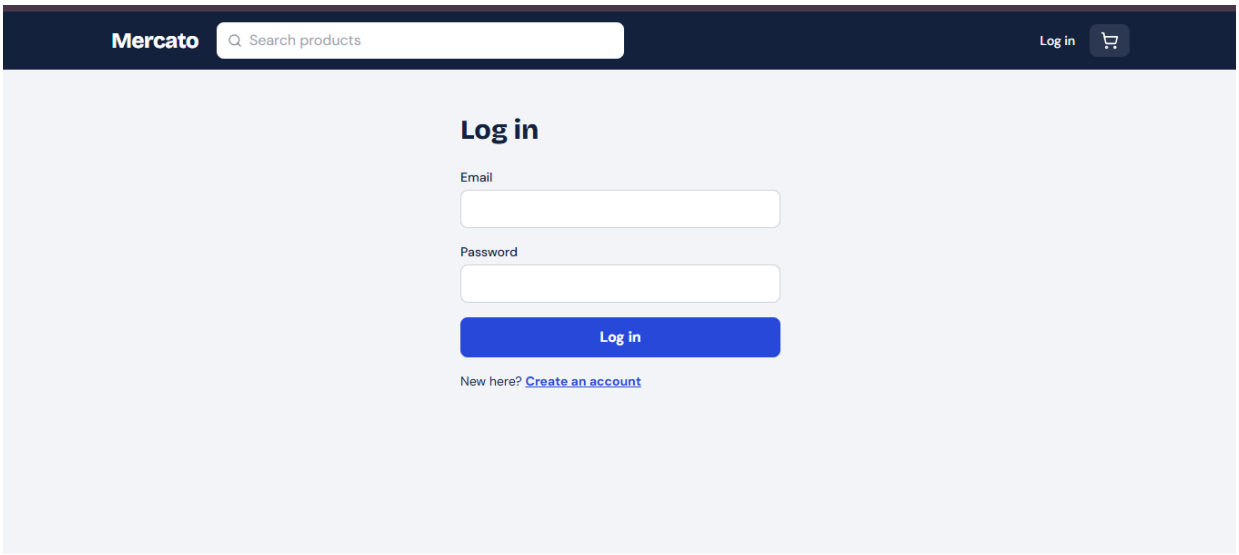
- Frontend: Vercel, root directory `frontend`
- Backend: Render web service, root directory `backend`
- Database: Aiven for MySQL (free plan, SSL on)

Screenshots

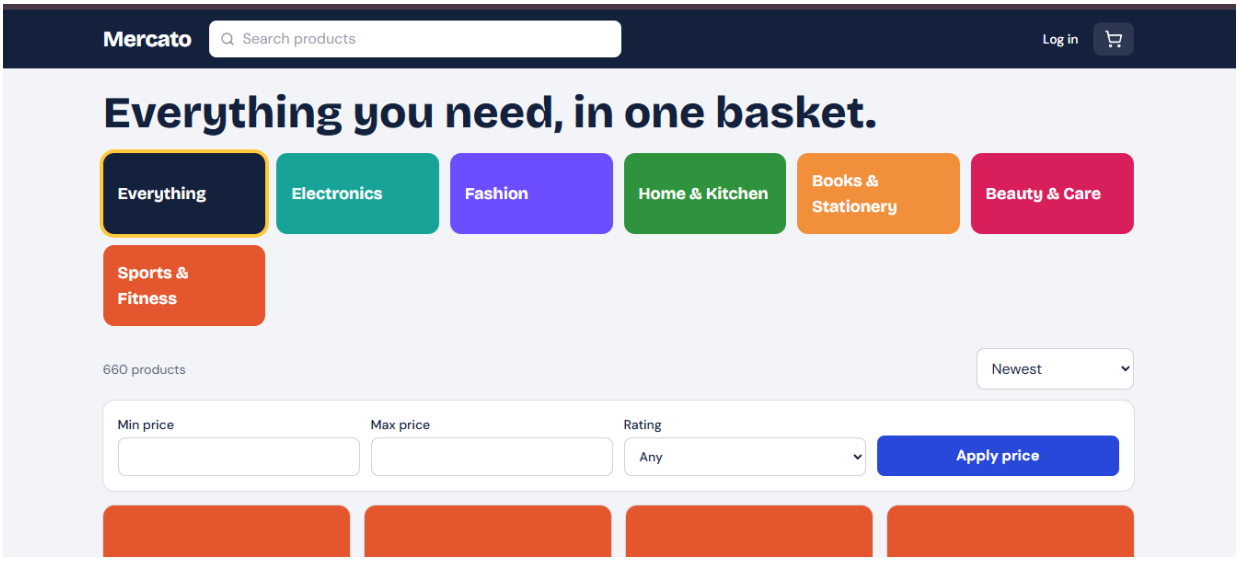
Home page :



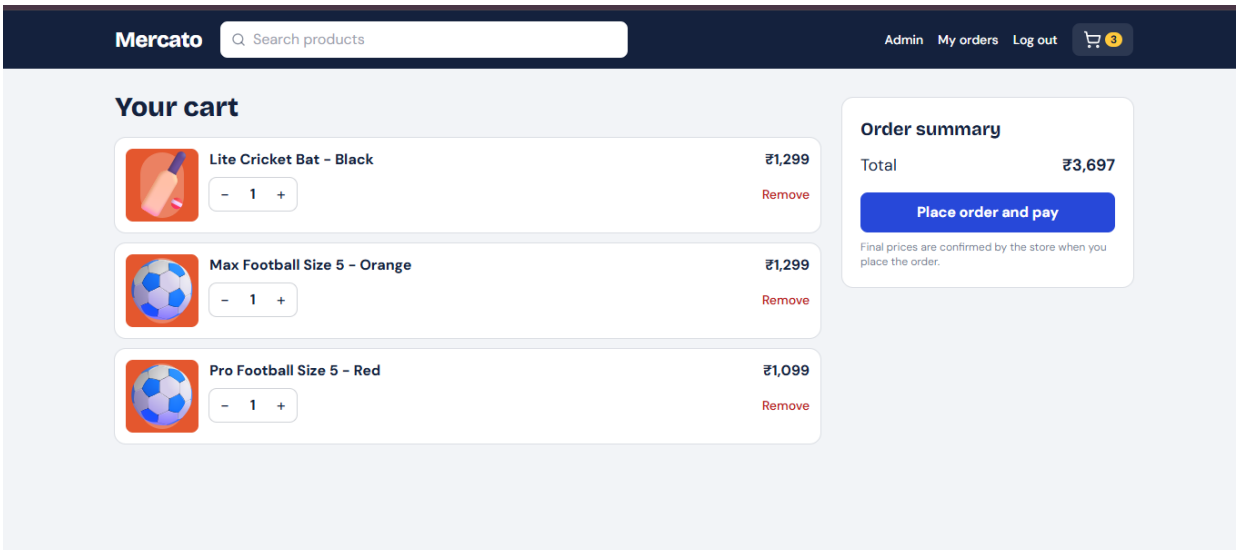
Login page :



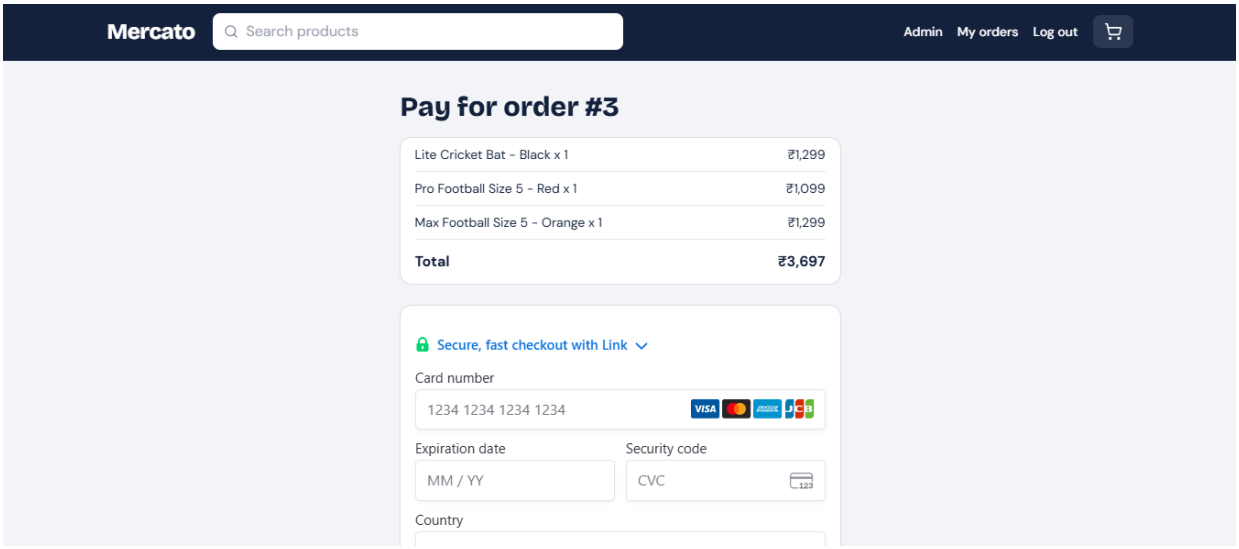
Product filters:

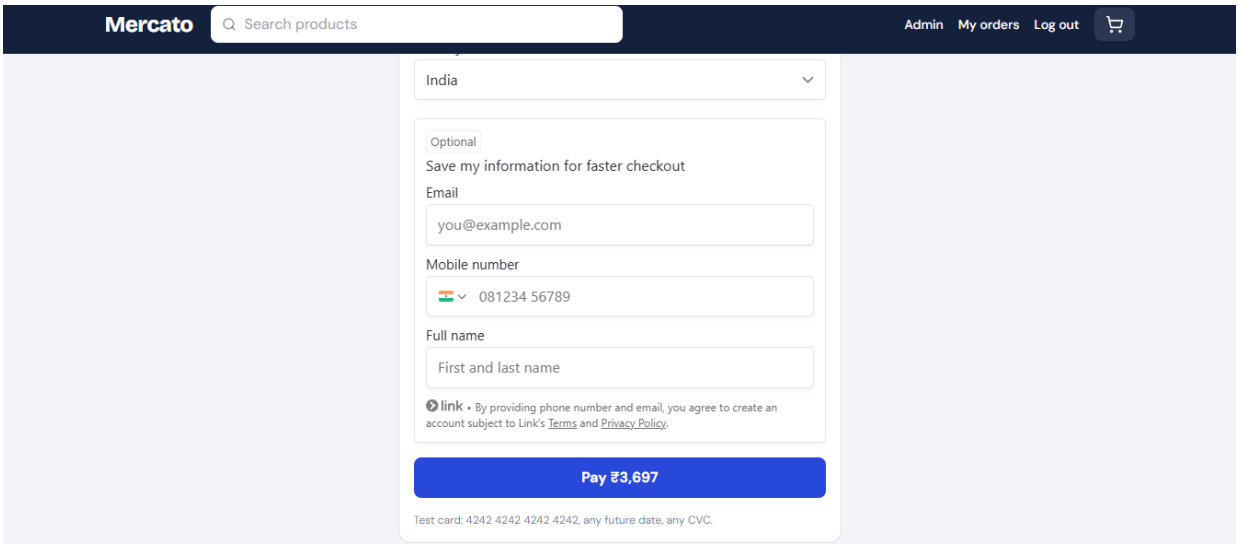


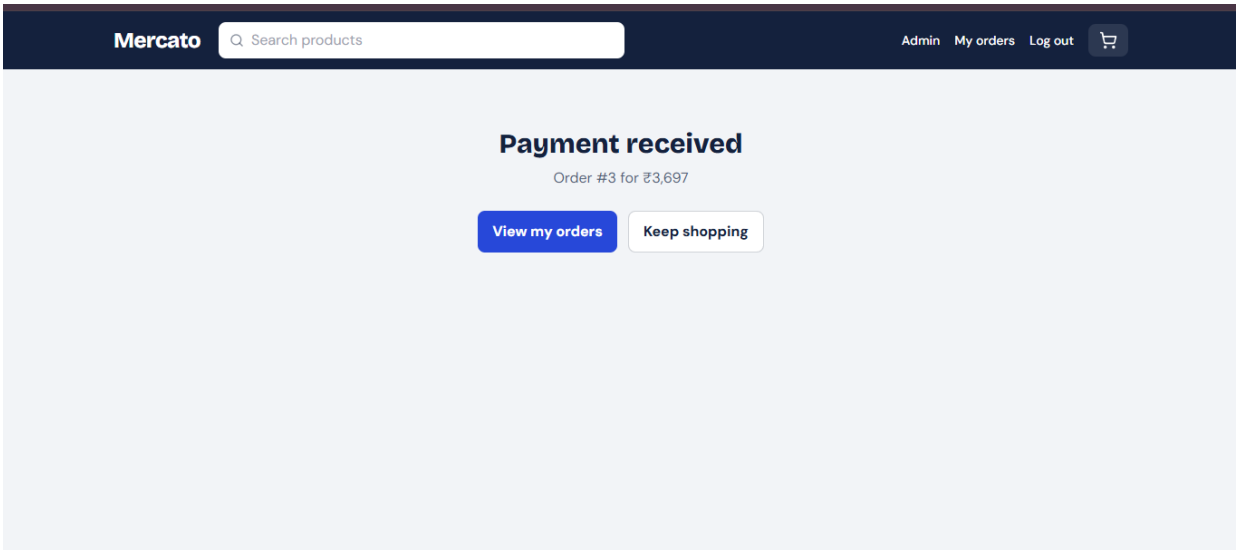
Cart :



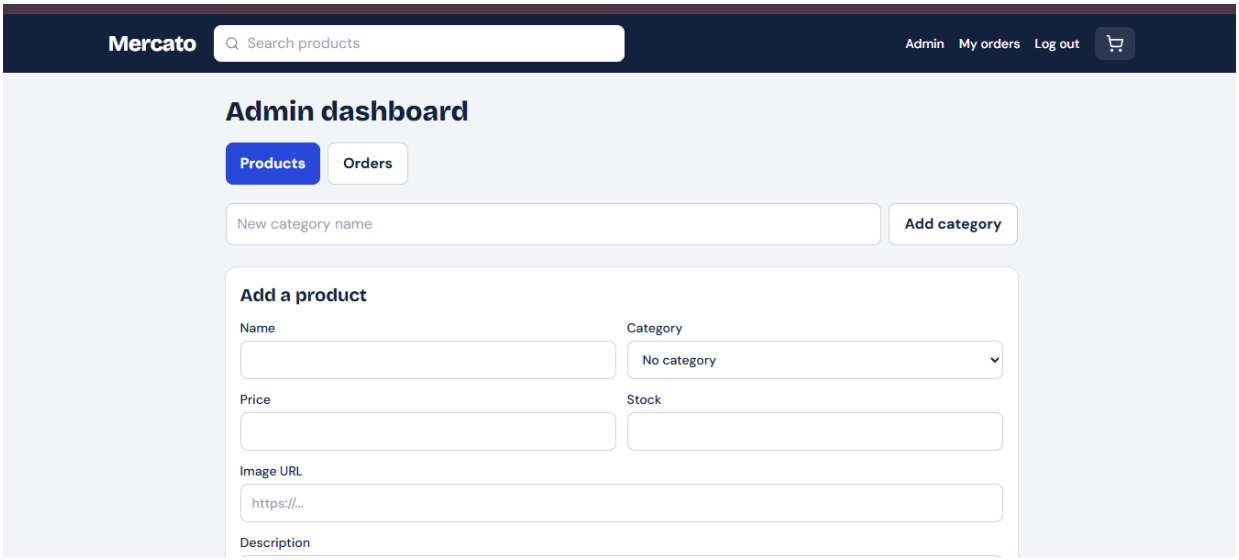
Payment success :

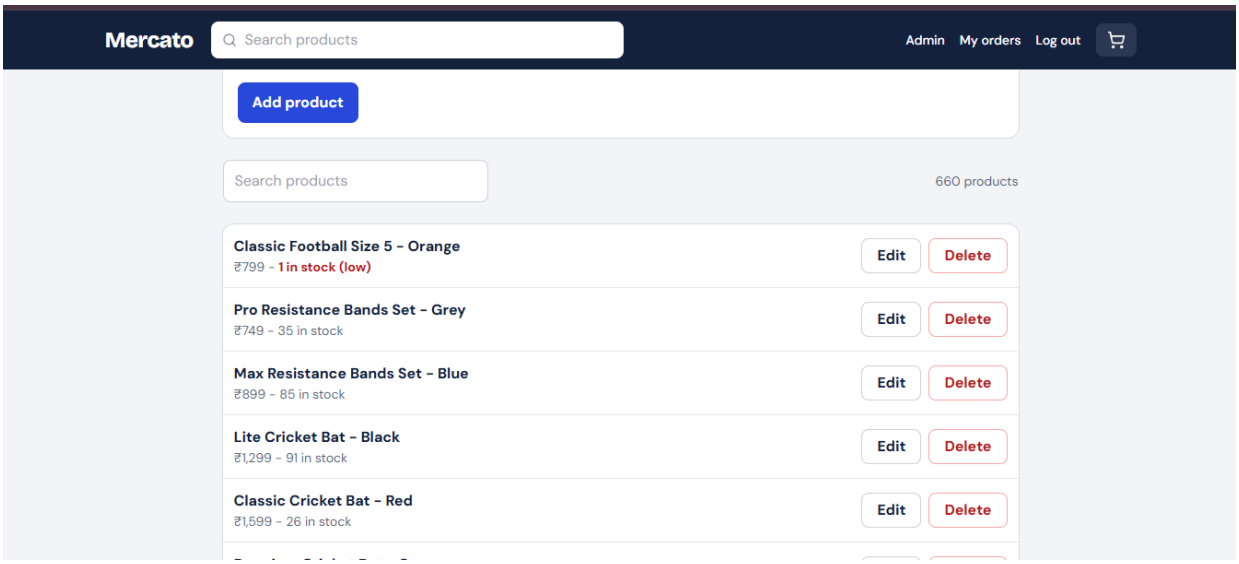






Admin dashboard





Author :

Built by [@smukeshkumardev](https://github.com/smukeshkumardev)